Family 1: grouped graph atlas

Exact reachable sizes 1 through 20

Each vertex has two outgoing edges and unrestricted indegree. Loops and parallel edges are allowed. Arrows have identical styles: their local port order is immaterial. Vertex numbers identify possible starting states.

For a chosen start, $H(G)$ is the shortest positive length of two distinct equal-length words whose endpoint vectors agree for every static mask. Different starting states are scored separately. A double circle marks an alternative start, never a simultaneous collection of roots.

m	$F(m)$	Rooted winners	Shapes	Evidence
1	1	1	1	Certified complete
2	3	1	1	Certified complete
3	6	2	1	Certified complete
4	10	1	1	Certified complete
5	15	5	4	Certified complete
6	23	2	1	Certified complete
7	34	8	4	Certified complete
8	≥ 35	–	–	Global maximum unresolved
9	≥ 34	–	–	Constructive lower example
10	≥ 34	–	–	Constructive lower example
11	≥ 34	–	–	Constructive lower example
12	≥ 34	–	–	Constructive lower example
13	≥ 34	–	–	Constructive lower example
14	≥ 34	–	–	Constructive lower example
15	≥ 34	–	–	Constructive lower example
16	≥ 34	–	–	Constructive lower example
17	≥ 34	–	–	Constructive lower example
18	≥ 34	–	–	Constructive lower example
19	≥ 34	–	–	Constructive lower example
20	≥ 34	–	–	Constructive lower example

For $m \leq 7$, the 20 certified maximizing rooted classes are grouped into 13 ordinary graph shapes. Every vertex giving one of those rooted classes is marked, including automorphically equivalent vertices. The source witnesses and full root-to-class mappings are preserved in the accompanying JSON.

At $m = 8$, the picture is the current heuristic leader: finder length 50, estimated $H \approx 49$ (rough range 48–50), rigorous graph bound $1 \leq H \leq 50$. The independently certified global lower bound $F(8) \geq 35$ comes from a different graph. No exact score or global maximality is claimed for the pictured graph.

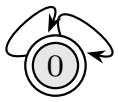
For $9 \leq m \leq 20$, a marked vertex is only the start of an example with $H \geq 34$, proved by projection to a certified seven-state graph. These are not asserted to be winners.

Exactly m reachable vertices is distinct from a cumulative budget $1, \ldots, m$. Deleting unreachable vertices preserves all rooted trajectories and $H(G)$; the cumulative optimum is therefore the maximum over the exact reachable sizes. No strong connectivity assumption is imposed.

$m = 1$

Global maximum $F(1) = 1$; all 1 rooted maximizing classes, grouped into 1 ordinary shape.

Double-circled vertices are all optimal starting states. Choose one marked start at a time; each choice defines a separate rooted graph.

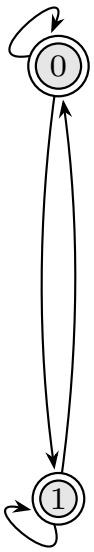


(a) $H = 1$. Optimal starts: $\{0\}$.
1 distinct rooted maximizing class.

$m = 2$

Global maximum $F(2) = 3$; all 1 rooted maximizing classes, grouped into 1 ordinary shape.

Double-circled vertices are all optimal starting states. Choose one marked start at a time; each choice defines a separate rooted graph.

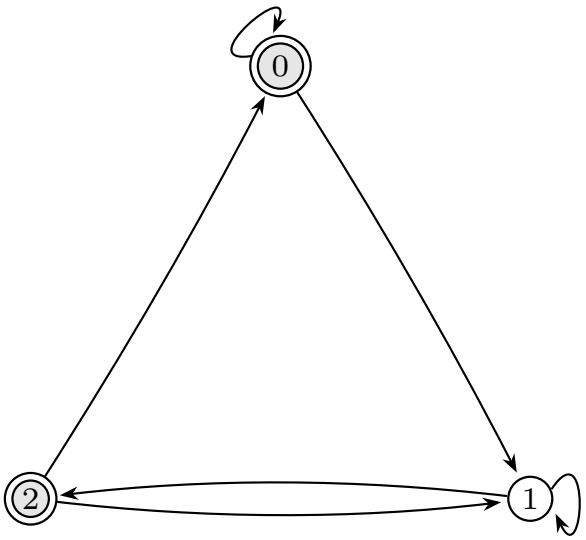


(a) $H = 3$. Optimal starts: $\{0, 1\}$.
1 distinct rooted maximizing class.

$m = 3$

Global maximum $F(3) = 6$; all 2 rooted maximizing classes, grouped into 1 ordinary shape.

Double-circled vertices are all optimal starting states. Choose one marked start at a time; each choice defines a separate rooted graph.

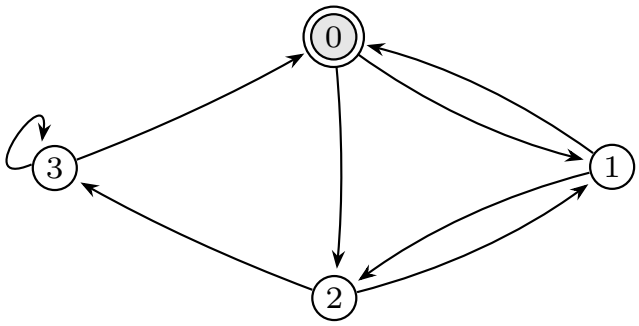


(a) $H = 6$. Optimal starts: $\{0, 2\}$.
2 distinct rooted maximizing classes.

$m = 4$

Global maximum $F(4) = 10$; all 1 rooted maximizing classes, grouped into 1 ordinary shape.

Double-circled vertices are all optimal starting states. Choose one marked start at a time; each choice defines a separate rooted graph.

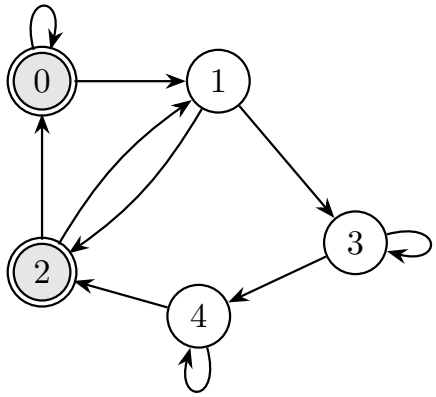


(a) $H = 10$. Optimal starts: $\{0\}$.
1 distinct rooted maximizing class.

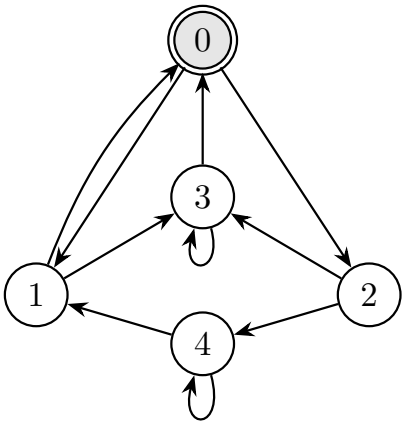
$m = 5$

Global maximum $F(5) = 15$; all 5 rooted maximizing classes, grouped into 4 ordinary shapes.

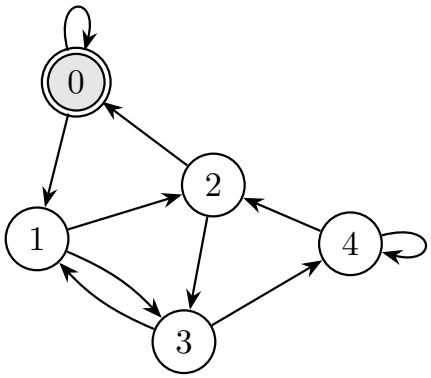
Double-circled vertices are all optimal starting states. Choose one marked start at a time; each choice defines a separate rooted graph.



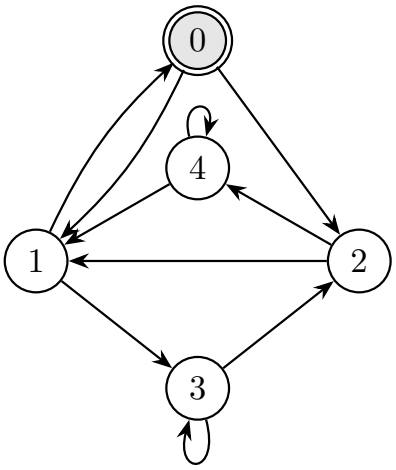
(a) $H = 15$. Optimal starts: $\{0, 2\}$.
2 distinct rooted maximizing classes.



(b) $H = 15$. Optimal starts: $\{0\}$.
1 distinct rooted maximizing class.



(c) $H = 15$. Optimal starts: $\{0\}$.
1 distinct rooted maximizing class.

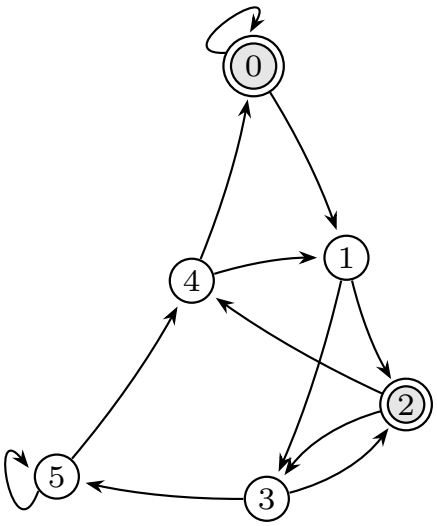


(d) $H = 15$. Optimal starts: $\{0\}$.
1 distinct rooted maximizing class.

$m = 6$

Global maximum $F(6) = 23$; all 2 rooted maximizing classes, grouped into 1 ordinary shape.

Double-circled vertices are all optimal starting states. Choose one marked start at a time; each choice defines a separate rooted graph.

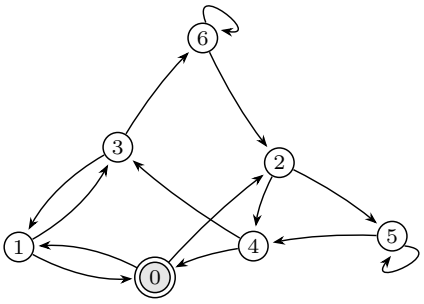


(a) $H = 23$. Optimal starts: $\{0, 2\}$.
2 distinct rooted maximizing classes.

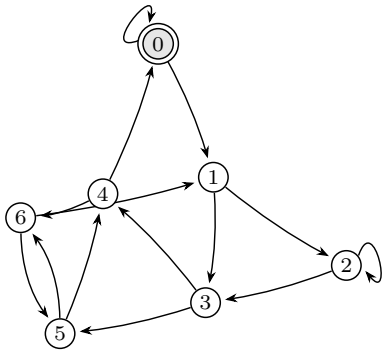
$m = 7$

Global maximum $F(7) = 34$; all 8 rooted maximizing classes, grouped into 4 ordinary shapes.

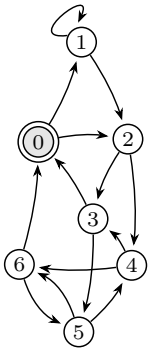
Double-circled vertices are all optimal starting states. Choose one marked start at a time; each choice defines a separate rooted graph.



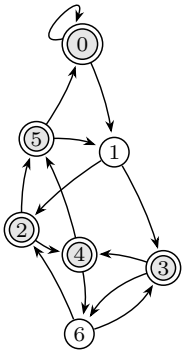
(a) $H = 34$. Optimal starts: $\{0\}$.
1 distinct rooted maximizing class.



(b) $H = 34$. Optimal starts: $\{0\}$.
1 distinct rooted maximizing class.



(c) $H = 34$. Optimal starts: $\{0\}$.
1 distinct rooted maximizing class.



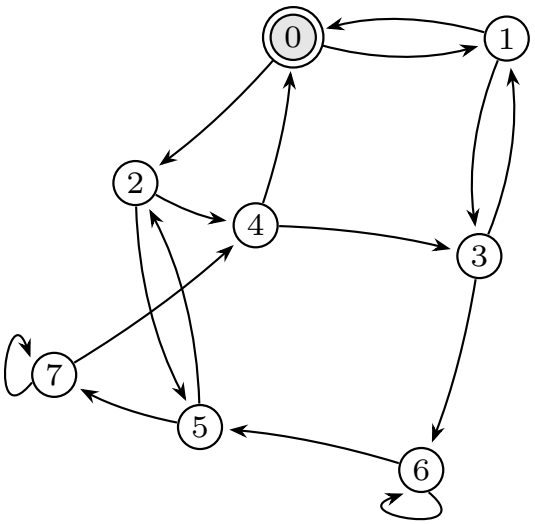
(d) $H = 34$. Optimal starts: $\{0, 2, 3, 4, 5\}$.
5 distinct rooted maximizing classes.

$m = 8$

One candidate found. The working estimate is 49 (rough range 48–50), with a verified pair of length 50. The marked start 0 gives the best current estimate; start 2 of the same graph is the runner-up.

All eight starting states of this graph were checked heuristically. The returned pair lengths are:

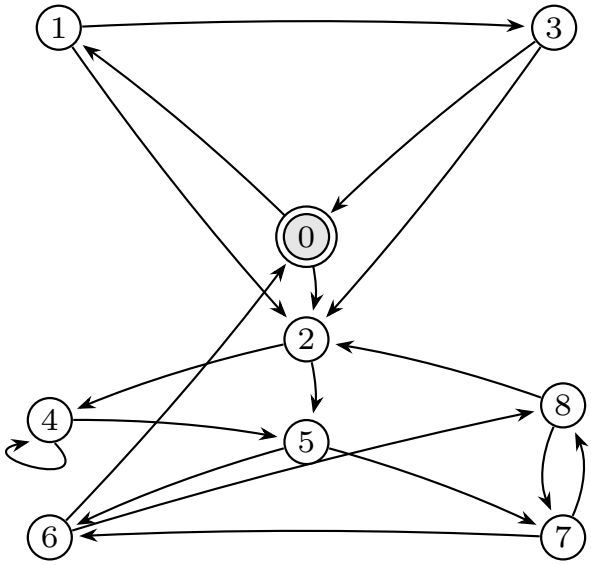
Start	0	1	2	3	4	5	6	7
Pair length	50	42	49	39	43	47	43	46



(a) Best current estimated start: 0.
Finder length 50; working estimate $H \approx 49$.

$m = 9$

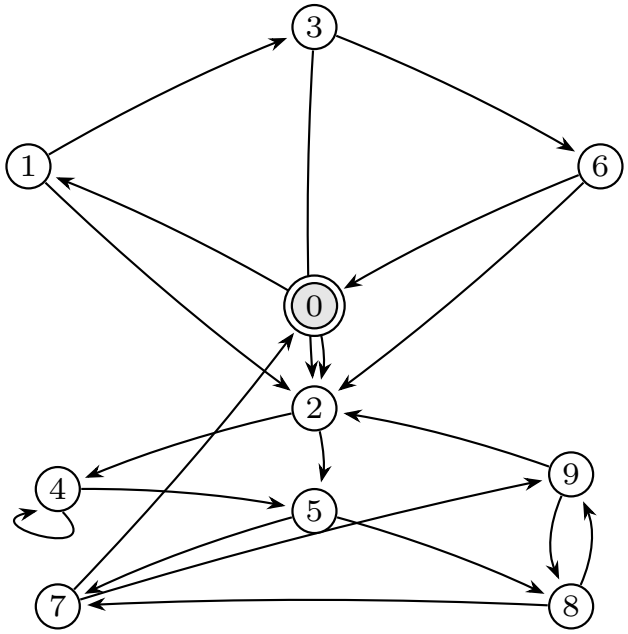
Constructive lower-bound example. The indicated start gives $H \geq 34$ by a static-mask projection onto a certified seven-state maximizer. This graph has not been certified globally optimal; its exact score is unresolved.



(a) Example start: 0. Proved $H \geq 34$.
Exact score and global maximum unresolved.

$m = 10$

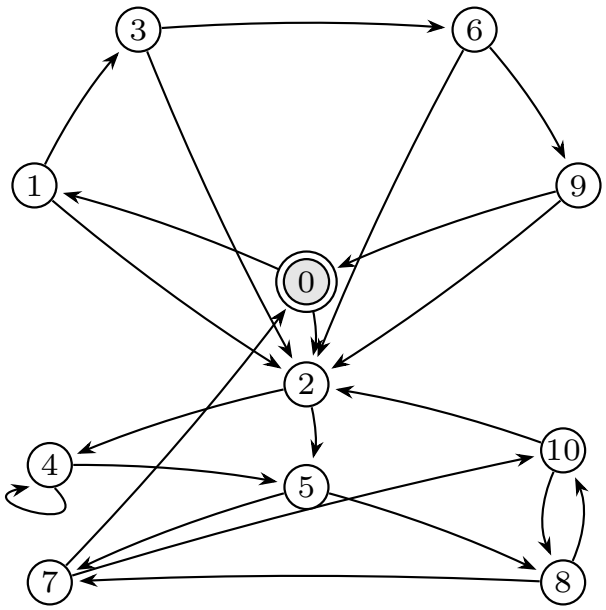
Constructive lower-bound example. The indicated start gives $H \geq 34$ by a static-mask projection onto a certified seven-state maximizer. This graph has not been certified globally optimal; its exact score is unresolved.



(a) Example start: 0. Proved $H \geq 34$.
Exact score and global maximum unresolved.

$m = 11$

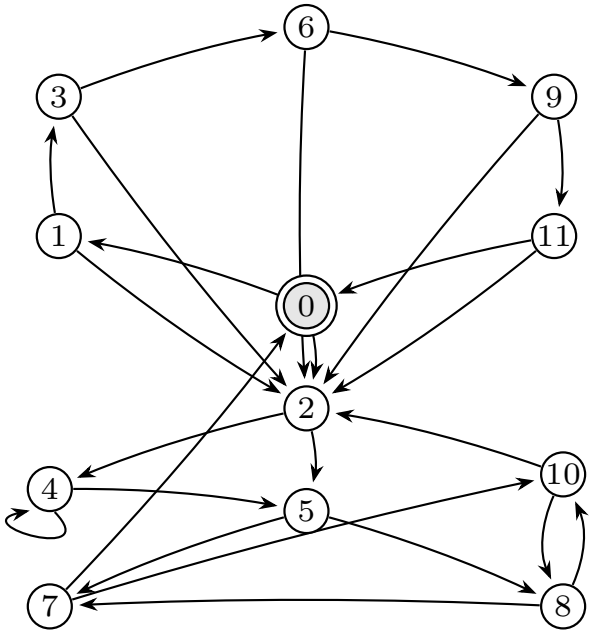
Constructive lower-bound example. The indicated start gives $H \geq 34$ by a static-mask projection onto a certified seven-state maximizer. This graph has not been certified globally optimal; its exact score is unresolved.



(a) Example start: 0. Proved $H \geq 34$.
Exact score and global maximum unresolved.

$m = 12$

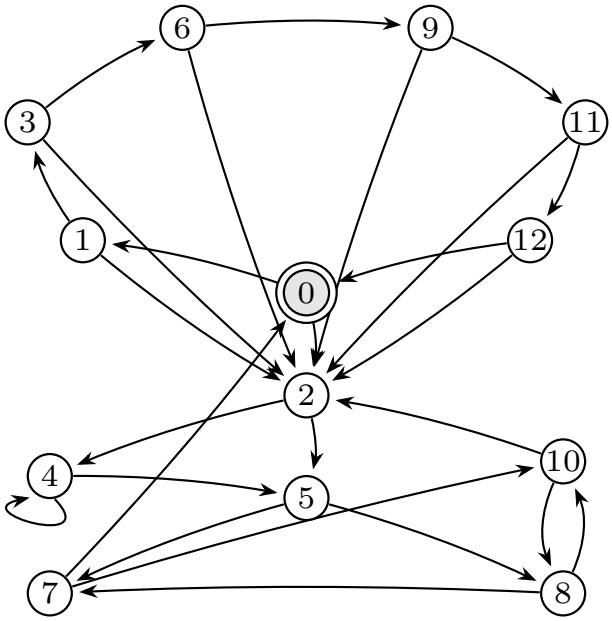
Constructive lower-bound example. The indicated start gives $H \geq 34$ by a static-mask projection onto a certified seven-state maximizer. This graph has not been certified globally optimal; its exact score is unresolved.



(a) Example start: 0. Proved $H \geq 34$.
Exact score and global maximum unresolved.

$m = 13$

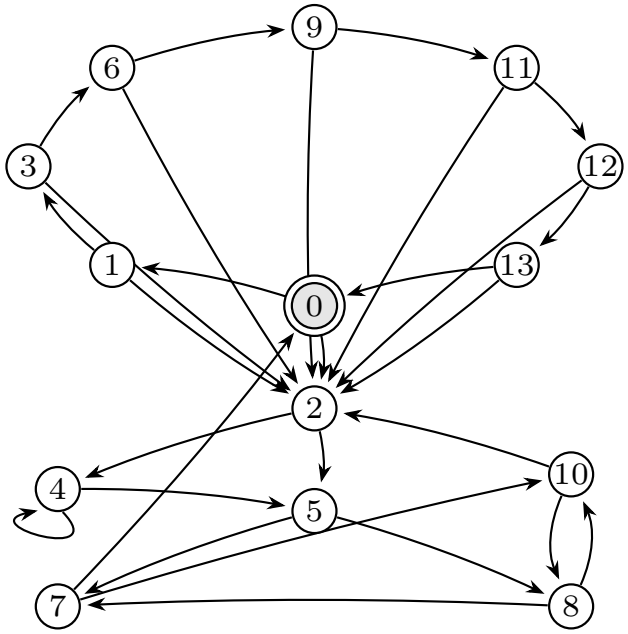
Constructive lower-bound example. The indicated start gives $H \geq 34$ by a static-mask projection onto a certified seven-state maximizer. This graph has not been certified globally optimal; its exact score is unresolved.



(a) Example start: 0. Proved $H \geq 34$.
Exact score and global maximum unresolved.

$m = 14$

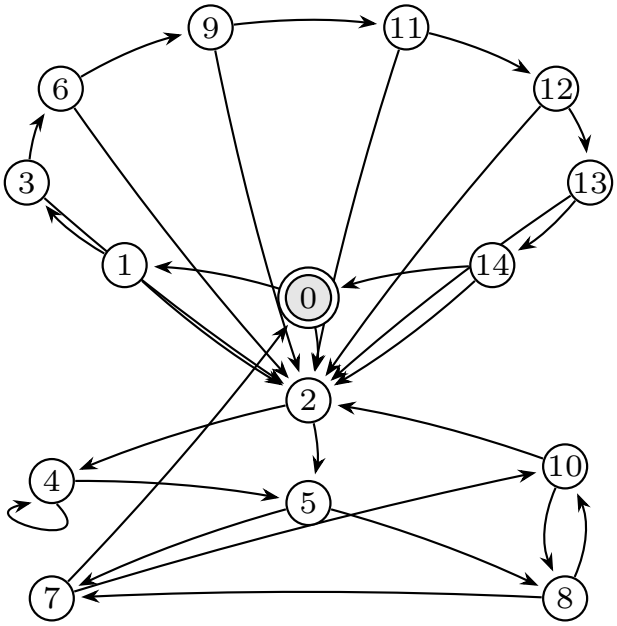
Constructive lower-bound example. The indicated start gives $H \geq 34$ by a static-mask projection onto a certified seven-state maximizer. This graph has not been certified globally optimal; its exact score is unresolved.



(a) Example start: 0. Proved $H \geq 34$.
Exact score and global maximum unresolved.

$m = 15$

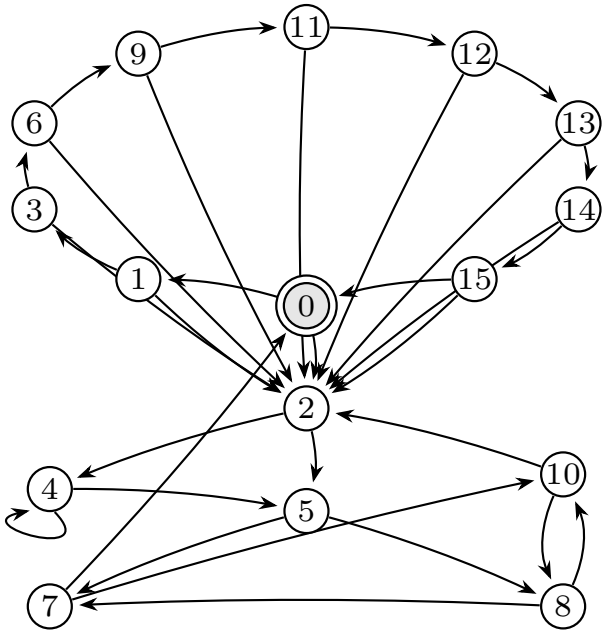
Constructive lower-bound example. The indicated start gives $H \geq 34$ by a static-mask projection onto a certified seven-state maximizer. This graph has not been certified globally optimal; its exact score is unresolved.



(a) Example start: 0. Proved $H \geq 34$.
Exact score and global maximum unresolved.

$m = 16$

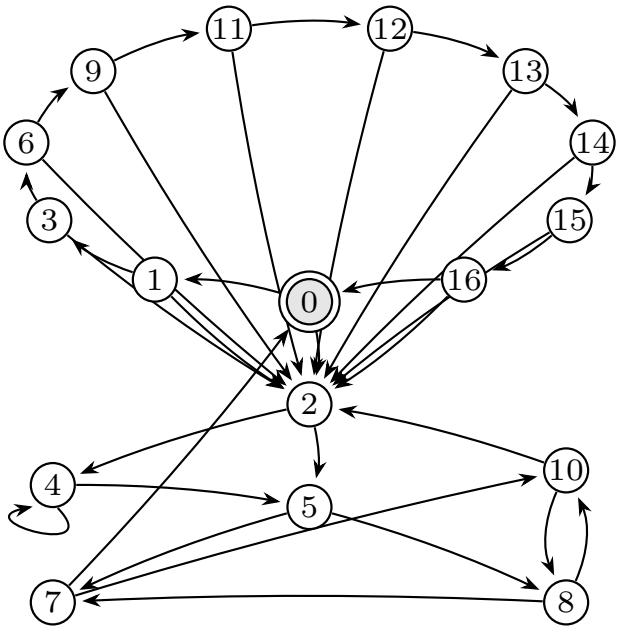
Constructive lower-bound example. The indicated start gives $H \geq 34$ by a static-mask projection onto a certified seven-state maximizer. This graph has not been certified globally optimal; its exact score is unresolved.



(a) Example start: 0. Proved $H \geq 34$.
Exact score and global maximum unresolved.

$m = 17$

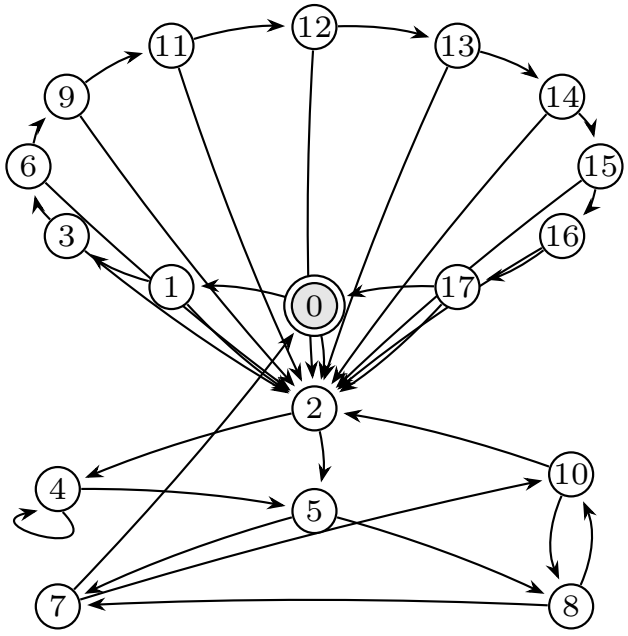
Constructive lower-bound example. The indicated start gives $H \geq 34$ by a static-mask projection onto a certified seven-state maximizer. This graph has not been certified globally optimal; its exact score is unresolved.



(a) Example start: 0. Proved $H \geq 34$.
Exact score and global maximum unresolved.

$m = 18$

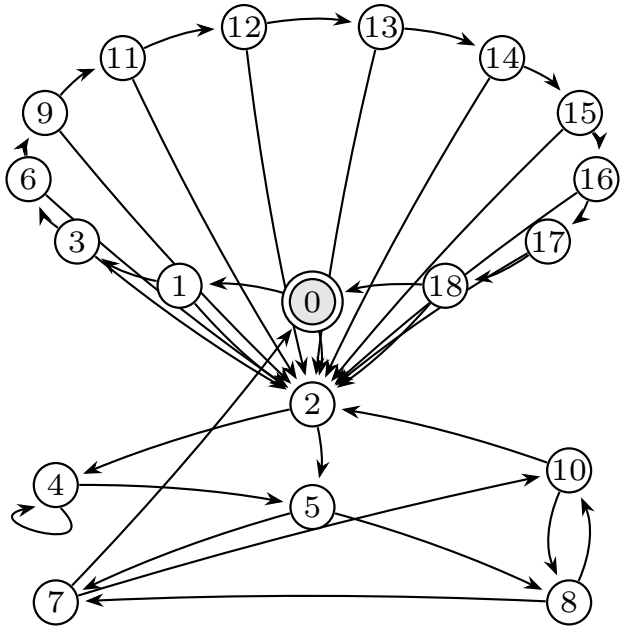
Constructive lower-bound example. The indicated start gives $H \geq 34$ by a static-mask projection onto a certified seven-state maximizer. This graph has not been certified globally optimal; its exact score is unresolved.



(a) Example start: 0. Proved $H \geq 34$.
Exact score and global maximum unresolved.

$m = 19$

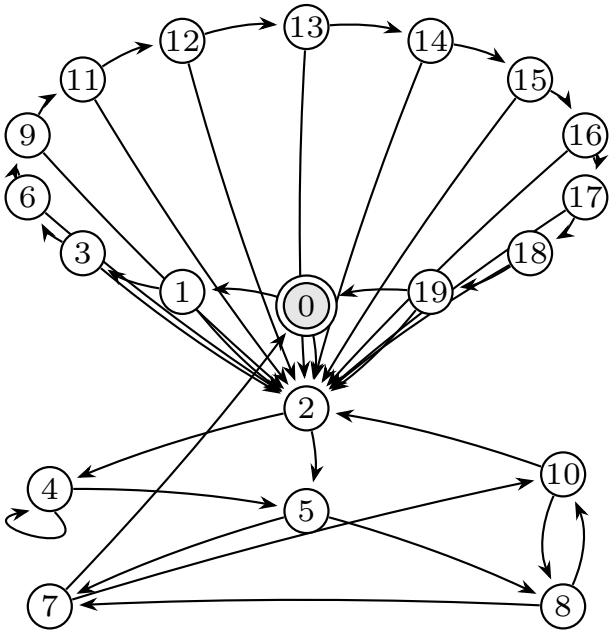
Constructive lower-bound example. The indicated start gives $H \geq 34$ by a static-mask projection onto a certified seven-state maximizer. This graph has not been certified globally optimal; its exact score is unresolved.



(a) Example start: 0. Proved $H \geq 34$.
Exact score and global maximum unresolved.

$m = 20$

Constructive lower-bound example. The indicated start gives $H \geq 34$ by a static-mask projection onto a certified seven-state maximizer. This graph has not been certified globally optimal; its exact score is unresolved.



(a) Example start: 0. Proved $H \geq 34$.
Exact score and global maximum unresolved.